Authorial Burden

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Abstract Limits that emerge out of the interactive nature of interactive digital narrative make authoring it challenging. These limits include exponential branching, where branches in the narrative increase the amount of content needed to be written progressively throughout the work; combinatorial explosion, where increasing combinations of possible game states makes writing additional content complex, as well as programming scope problems that are seen in any digital project, wherein the range of features or game interactions that could be implemented is infinite but development time finite. These limits place on the authors of interactive digital narrative an authorial burden, increasing the amount of content needed to be written, states managed or features programmed. Multiple strategies exist for tackling the burden, from reducing or reusing content, to decontextualising and generating content.

1 Introduction

Any creative endeavour has its difficulties, many of which arise out of their respective mediums. Painters mix paints; filmmakers must control the lighting in their shots. Interactive narrative similarly has its own creative challenges. Allowing interaction—especially choices which lead to alternative content—can require a great deal of additional content to be created. This content can be increasingly complex to the author. This has been referred to as the authorial burden.[1] This isn't meant to be pejorative: the act of writing doesn't need to feel burdensome. Rather, the authorial burden refers to the workload cost of authoring incurred by making interactive design decisions.

[1]The first appearance of this term appears to be by Mateas and Sterne in 2002 when discussing the authoring of their story-game, *Façade*, [38] though the problem clearly predates the term. This term has appeared regularly in the literature since then [16, 19, 28, 42, 44, 49, 50, 53–56, 62].

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Content can be slow to write because there is a great deal of it to produce, or it can be slow to write because each new bit of content is complex to the author: Garbe refers to these limits as the 'authoring wall' and the 'complexity ceiling' [22]. Authoring interactive digital narrative is not just writing, but rather a range of complimentary activities including planning, visualisation, structuring, testing and so forth [34]. If content is difficult to visualise or test, then it will be slower to write and have a greater concomitant burden.

1.1 Authoring Wall

If you have a branching story with two equal length paths, then you'll need to write twice as much; if you have a story with content displayed differently based on combinations of different game states, then adding more states exponentially increases the content required; if you give the player a great deal of freedom of action, i.e. if the scope is particularly large, then a greater amount of content needs to be written.

The simplest structure of interactive narrative is the branching story, in which choices split the plot and the separate paths do not converge. This tree structure is sometimes known as the broomstick (especially with regards to endings) [8] or the Time Cave (after the Choose-Your-Own-Adventure novel of the same name that had this structure) [2]. Time Caves most quickly run into the problem of exponential branching. Without merging branches, any kind of branching work faces the limit of exponential writing. To date, the longest complete branching work which has no merging is *Girth Loinhammer's Exponential Adventure* [63]. This is only ten passages long, with each complete playthrough having nine choices with exactly two options each. This leads to 512 separate endings. If the author, Damon Wakes, were to write one extra choice for every possible playthrough, he would have to write 1024 additional passages. Another choice on top of that would be 2048 additional passages. With this simple structure, the authorial burden doubles every time the average game length is increased by one choice. The limit on how long a narrative of this type can be is quickly reached.

1.2 Complexity Ceiling

The authorial burden can be large because of the amount of content that needs to be written, but it also can be large because the content that needs to be authored is difficult to write due to state complexity. This can take the form of multiple different plot-states that need to be kept in mind while authoring, or trying to author snippets of content where the final presentation is unclear, or attempting to account for combinations of states. Scenes become more difficult to write the more alternative states there are. For example, an author might have to take into account whether two

characters in a scene had already met, or what they know about each other, and if they had already met what happened between them.

Classic authoring problems from literature, like avoiding writing oneself into a corner, or maintaining continuity in a long work are compounded in an interactive piece if multiple possible plot-states can be true for any given scene, or if certain choices must be disallowed to ensure continuity later on in the story. For instance, a character dying can incur an authoring burden for future scenes in which that character could appear, requiring extensive additional writing; and the act of making sure that any given scene isn't incurring an undue burden is itself an additional factor that increased can slow down authorship.

The simple branching structure has only one game state at any given time. A common way to manage choice in interactive fiction is through tracking multiple states. This allows any choice to have an effect on branching the story without requiring that every choice immediately creates a hard fork. However, works that rely on combinations of states run into a different problem of unmanageable combination sets. When each new state added can be combined with a number of existing states and new content needs to be written for this combination, the author eventually runs into a limit of complexity that they can handle [23].

1.3 Forms

Interactive narrative can take many different forms with varying authorial burdens. Long-form branching narrative, whether text-based, filmed or animated, where the player selects from choices, will be created differently to classic text adventure with a parser where players type commands. Compare this with agent-based simulationist works where a narrative emerges from interaction with virtual actors requires different skills, or database fiction where a player explores some body of content in order to make their own connections, or games like walking sims where the player explores a 3D environment. Some narratives require tactical decision-making from the player or puzzle solving, others have purely kinetic interaction. Nevertheless, the underlying interactive nature of these various forms generates a need for authors to write additional content which, if not properly managed, can expand exponentially. As such, many of the broad approaches for managing the authorial burden apply to multiple different forms of interactive narrative and many of these different forms can themselves be understood not only as aesthetic decisions but also as means of overcoming the constraints associated with authoring content which varies.

The intended length of experience is intertwined with the chosen form. Some works are intended to be played through once to completion, others are designed as repeatable experiences. Of those repeatable experiences, some might allow short iterative playthroughs, others (like *Fallen London* [20]) may be designed to come back to repeatedly over long periods of time and perhaps never reach an ending. These different intended lengths have their own concomitant burden; a work which

is designed to be experienced once may employ approaches that wouldn't be as satisfying in a work designed for repeat experience.

The form of a digital work a player may experience doesn't necessarily reflect its coded structure. A well-written loop with expansive use of textual variation might be experienced by a player as a linear series of similar but distinct events. A branching section can be experienced as linear by a player who is unaware that variations were possible. It is the creative decision about the work-as-written rather than the work-as-experienced that incur an authorial burden.

1.4 Strategies for Unburdening

There are strategies for reducing the authorial burden that can be deployed to reduce or re-use content, decouple or generate segments of content, or otherwise improve the process of authorship.

i **Reducing** Authors can seek to cut the authorial burden by reducing the amount of possible content, such as by limiting scope or merging branches.
ii **Reusing** Existing content (such as backgrounds, animations and even scenes) can be repurposed, thereby avoiding the need to create additional content. The same representation can be repeated (e.g. having the same scene occur in two branches) or the same assets, text snippets or animations may be reused to represent different things (e.g. using the same library of responses for different non-player characters).
iii **Decoupling** New content can be harder to write the more it needs to be continuous with existing content. Making sections more context-independent can reduce the writing complexity, allowing more to be written. Structuring works into distinct episodes can be a form of decoupling.
iv **Generating** Content can be procedurally generated, allowing a great deal of possible novel combinations. This may reduce the writing burden while increasing other forms of workload (e.g. testing and programming).
v **Embracing** Rather than changing the form of an interactive narrative itself, the ability of authors to write can be improved. These approaches embrace the authorial burden by increasing the capacity of authors to tackle it.

These kinds of strategies can be seen instantiated in the various approaches to tackling the authorial burden. It should be noted that there are other reasons for making design choices beyond their effect on the total workload. Managing authorial labour is only one of several reasons an author might use a specific structure of interaction or set of tools. Different plot types, such as the epic plot-based or the dramatic character-based form may fit better or worse with different modes of interaction [57]. Narrative structures expressed through choices offered to the player, or actions possible, are also poetic gameplay devices for evoking specific emotions in the player [10]. Interactivity is deployed for different purposes, depending on the experience

the creators intend. Interactive elements can have multiple functions, even within the same work. For instance, puzzles might be included for challenge, humour, or to gate progress based on the player's understanding of the story. A loop might be present to establish a theme of repetition or return. Clickable text links may be used for pacing, or to develop a sense of complicity.

Nevertheless, given the untenability of compounded branching, any story with alternative content will necessarily be authored with some strategy for managing the authorial burden. Strategies aren't mutually exclusive, and the categories can overlap. Reusing content is at the same time reducing possible content.

2 Reducing

The most straightforward way that a digital project can involve a great deal of writing is by having a large scope. Indeed, one way of conceiving of the authorial burden is as under-managed scope. All completed projects necessarily limit scope to some extent: decisions always have to be made about what not to include. Scope limitation comes in different forms. In interactive narrative, scope is limited along at least these two axes: objects and verbs.

2.1 Limiting Objects

Implementing a large range of objects in a digital work (locations, characters, items, musical scores, etc.) incurs a burden of content creation for all things implemented. A typical way of reducing scope is to simply cut planned or possible content. In multimedia (animation, acting, illustration, etc.), keeping the range of assets used manageable can be important for keeping costs down and not just for time reasons. This has been called the 'economics of building' [47]. Diagetic constraints are often employed, grounding the limitation of elements in the fiction: aside from reasons of plot and atmosphere, this is why there is a plethora of closed houses, isolated islands and small casts in interactive narrative. As in a novel, in a purely textual work with minimal state-tracking and no visual assets, new locations and characters do not pose the same limitations.

2.2 Limiting Verbs

In an interactive work, the player is typically able to undertake various actions. This can range from choosing between links to click, all the way up to simulating a virtual environment. The range of verbs allowed then might range from one to many hundred (in the case of some parser-based games). For these games where the player

might type anything, some authors have chosen to supply a very curtailed list of accepted words, but then implement bespoke responses for every valid combination. CEJ Pacian employs this approach in several of his works and calls the approach 'shallow but broad' [46]: in *Superliminal Vagrant Twin*, [46] the player can travel to a large range of planets, allowing a great sense of openness, but this is achieved by tightly limiting the possibility space on each planet to less than a dozen options.

2.3 Primacy of Text or Dialogue

Assets range considerably in cost and development time. Next to animation, live filming, illustration and so forth, text is comparatively cheap. Works can contain a great deal of text in descriptions and dialogue in a highly scalable way, in that adding more lines of text does not typically require new capabilities. This primacy isn't absolute: while a line of unique dialogue may typically appear once in a work, the experience of many games is made up of performing repeatable actions in a virtual space.

Similarly, lines of dialogue don't necessarily require unique animations or assets (though in some productions they may be voiced), and so by centring the exchange of words as the primary experience, the authors save on having to produce and program animation that would be required in a more physically expressed experience. This is a common strategy in computer roleplaying games which can have tens of thousands of lines of dialogue [60].

2.4 Existing Tools

Creating bespoke engines is programming labour which can have a huge time cost, and so using existing narrative engines, programming languages, visualising tools, etc. can save a lot of possible work, with the compromise that what can be made will be limited to what is possible within those tools. For example, the walking sim *Dear Esther* [52] was originally written as a modification for the Source game Engine [47]. This allowed the game to be made using a ready-to-use existing tool that the developers were familiar with, although until its remake, this limited what could be achieved both in the structure of the island and in the visual art style.

2.5 Abstraction

Abstracting away realist elements is a common scope management technique. Players of games in general, including narrative games, will accept a great deal of abstraction and repetition which might break the suspension of disbelief in film or literature; the

player can cultivate a playing stance which allows them to separate the gamelike elements from their understanding of the work's narrative [33]. For example, choice-based conversations will often happen outside of real time, or may be repeated word-for-word without breaking immersion. Players accept game conventions, such as not being able to scale trivial obstacles, or having numerical stats represent aspects of their protagonist, just as the theatre-goer accepts when stage-hands and the audience are ignored by the actors of a play.

2.6 Gauntlet

A classic solution to writing reasonably lengthy stories with choices but without compounded branching is the gauntlet structure [2]. In the gauntlet, the story has one core path with many short endings off of this path. In gamebooks, this would often be represented by premature deaths. This structure only suits certain kinds of stories, though it still can be seen in contemporary works such as in large parts of the interactive film Bandersnatch [7]. A variant of the gauntlet is the 'friendly gauntlet' in which there are no premature endings: all side branches fold back onto the main path (see Merging Branches below). Most episodes of the choice-based videogame *Life is Strange* [17] follow a friendly gauntlet structure.

3 Reusing

Reusing the same element (a scene, a piece of art, a sign-off in a conversation, etc.) can be economical. This reaches its limit if it becomes prohibitively difficult to write content using existing elements or to cohere with pre-decided plot points.

3.1 Merging Branches

Branching and then merging is the most fundamental strategy for managing interactive narrative. Across a whole work, it has been referred to as a 'branch and bottleneck' structure, for the way, paths can branch out in a story and then return at bottlenecks [2]. Hargood and Crawford have separately referred to this as a foldback [15, 27], in which the branches of the story bend back on themselves. Similarly, Bernstein refers to this pattern as the Split/Join [4]. Most other hypertext patterns are versions of this at different scales. This can be conceived of as a way to reduce possible content by sending the player back to a central trunk of content.

Branches can be merged instantly or after divergent content. The breadth and form of divergent content can vary from an inconsequential minor variation, to fully developed parallel branches.

3.1.1 Empty Choice

The purest merge is to offer a choice and redirect the player to the same subsequent content regardless of their pick, with no states tracked. Here the choice might be functioning as a pacing device or to offer the player a different sense of the story or protagonist. This has little concomitant burden beyond the labour involved in conceiving of the non-choices. This technique has been called Illusory Agency, [19], and the offering of False Choices [39]. Such a strategy may genuinely save labour, but players (especially on replay) may see through the device, especially if consistently or overwhelmingly employed.

3.1.2 Recognised Choice

A player's choice can be recognised with a small divergence such as some differing text or dialogue. This creates some writing burden, but the burden isn't compounded as the story continues the same regardless of what is chosen. Mawhorter et al. call this a Flavor Choice [39].

3.1.3 State Changes

A choice may have a delayed effect, [39] with outcomes tracked by the narrative. A relationship might change, or an item may be gained, etc. These states can then be used later in the narrative without the requirement to branch immediately. Choice of Games, a publisher of interactive novels, refers to this as Delayed Branching [18].

3.1.4 Parallel Branches

A more authorially taxing form of splitting and merging branches is having substantial branches run in parallel, which then might merge at choke-points within the story. Ashwell refers to this structure as the Quest, [2] and it can be seen in works like *80 Days* [31] where multiple different parallel paths are possible for traversing the narrative at any given time.

3.2 Loops

Structurally, loops are a way of re-using the same content or set of choices offered. In a loop, the player is returned to a point in a scene or location which they have experienced before. Large narrative games like *Fallen London* make extensive use of this structure, with most content capable of being re-experienced. Time travel stories such as *Elsinore* [24] are based around this core idea, allowing large swathes of the

game to be re-experienced, often with a separate player agenda on replays, allowing for continual agency despite repeated content [43].

3.2.1 Hub Nodes

The hub node is one use of a loop. It is a node of an interactive story which can be returned repeatedly in a cycle pattern until some condition is met to move forward [4]. These hubs can appear in an otherwise unidirectional structure. At a hub, a list of choices are offered. After exploring one of these choices (which may have its own sub-choices), the player returns to the hub. Each time they return, used choices may be removed and new choices may be added. This is often the way conversation systems are implemented. Clusters of hub nodes can be used to implement a location structure in a choice-based work that is more commonly seen in parser fiction (see for instance, *16 Ways To Kill A Vampire At McDonalds* [13], or *With Those We Love Alive* [9]). The hub node is an effective way of structuring narrative segments in which there are multiple smaller scenes that could coherently be experienced in any order.

3.3 World Modelling

Many of the strategies discussed are based on a choice branching structure. Moving away from this going further than the loop is implementing a world model of persistent locations, objects and characters. Text adventure games are distinguished by their use of world models. As models become more richly implemented, they create their own authorial burden (for instance, in implementing different verbs and accounting for various combinations of objects). Still, modelling a persistent set of locations and objects is very common in fully illustrated works, as creating artwork for a location is costly, so it often makes sense to re-use places as much as possible (this is a common feature of narrative adventure games, such as *The Secret of Monkey Island*, [35] where the same locations and characters are returned to repeatedly).

3.4 Cumulative Variables

A straightforward way of lowering complexity in tracking narrative states is to use cumulative variables instead of, say, multiple Boolean values. For example, the interactive space opera, *Mass Effect*, [6] tracks how 'Renegade' the protagonist is. Instead of checking a list of every possible time they acted in a renegade way, the player's Renegade score accumulates at such occasions, and when relevant the single variable is checked. This principle is used extensively in the works of Choice of Games [30].

4 Decoupling

Decoupling is a way of reusing content, or rather by allowing scenes to appear regardless of previous content. The more self-contained segments of content are, the easier new sections can be written without having to write lots of variations depending on the world-state. This structure lends itself to narrative genres such as the picaresque in which the protagonist might embark on a series of loosely connected adventures. Conversely, agent-driven simulations and tightly plotted dramas may be harder to decouple.

4.1 Storylets

The storylet structure is a clear example of decoupling. The term was coined by the writers of the massive multi-million word browser-based text game, *Fallen London* [20] to refer to the chunks of content that can be experienced in many different possible orders. When these storylets are displayed based on conditional triggers, this is referred to as Quality-Based Narrative, [58] or Sculptural Hypertext [5]. Storylets in Fallen London work on a principle that has been referred to as the 'fires in the desert' approach: the writers create self-contained chunks of story (the well-lit 'fires'), leaving it up to the reader to infer the linkage between these chunks (the dark 'desert' between these bright spots of story) [21]. This strong context independence asks more from the reader than in traditional storytelling where events have much clearer causal links.

4.2 Modularisation

Modularising interactive narrative is to separate it into relatively self-contained segments. This is the same concept as 'levels' in videogames more broadly. To take an example: most long-form interactive novels published by Choice of Games[2] adhere somewhat to their stylebook, a set of guidelines for content and structure. The content guidelines ensure that content is in line with the values of the publishers (inclusive choices, no hate speech, etc.). The structural guidelines are a set of 'best practices for game management'. One of the guidelines is that works should be comprised of approximately ten vignettes that occur in sequential order. This ensures that the games are of sufficient length to meet their expected standards, but it also manages branching. It ensures that on a macro-scale across the whole interactive novel, no matter how many internal branches in a chapter, and no matter how any of them may end, they each must lead on to the next chapter in sequence. This modular design prevents the stories from sprawling out to an unmanageable number of parallel branches.

[2] Such as *Choice of Robots*, [25] *Créme de la Créme*, [48] *Trials of the Thief-Taker* [32] and so on.

Essentially this approach is a more formal application of the 'branch and bottleneck' structure, but in other works, modules might be even more independent.

4.3 Episodes

Full modularisation into distinct episodes can allow for episodic release in much the same manner as television shows, a technique used by Telltale Games for many of their narrative games. Episodic release has the benefit of allowing the creators to make and receive feedback from manageable chunks, as well as develop more of a prolonged event around the release of a work. Fully decoupled episodes, such as those that comprise *Kentucky Route Zero* [12] allow players to jump directly into any of the released episodes, as very few states are tracked between episodes. Brendan Patrick Hennessy's *Known Unknowns* [29] goes a step further, allowing players to jump into any chapter within any of the episodes, allowing scenes to be re-experienced in the same manner a reader might return to a favourite scene in a novel.

5 Generating

Generating content combines the approaches of reusing and modularising content, most often allowing the same elements to be recombined in a wide variety of contexts. Combining elements can happen at various different scales. It can happen at the level of the individual collection of pixels (as in procedural animation) or environmental elements such as rooms or objects, but also at the level of the sentence, word or letter. For words, tools like Tracery allow word lists to be recombined to create a huge variety of grammatical sentences according to simple iterative rules [11].

5.1 Procedural Content Generation

Procedural Generation is a strategy for creating content with many possible states instead of hand authoring unique state combinations. Sufficient content is required to populate generative lists and testing is needed to make sure the output is of sufficient quality. Narrative designer Cat Manning once quipped that procedural generation means 'generating twice the content in twice the time'.[3] Creating variations with the same quality and variety as equivalent hand-authored content may require different authoring skills but isn't necessarily less work.

[3] See https://twitter.com/catacalypto/status/1470893540964134913.

5.2 Supplementary Generation

Rather than generating core narrative content, supplementary elements can be generated. Reed has presented a system for reducing the authorial burden of writing by procedurally generating satellite sentences for pacing and context [50]. Ryan has demonstrated recombinant conversation generation for creating filler conversations [55]. Neither of these approaches produce content as satisfactory as bespoke-authored content, but they could be used to supplement such content. For instance, by giving variations of greetings and goodbyes and other formulaic conversation elements.

5.3 Natural Language Processing

Natural language processing is a set of tools for parsing user input. Rather than create bespoke responses to every possible useful input (as would be common in normal parser fiction), this approach seeks to dynamically 'understand' user input to decide what content should be shown next.

This technique is demonstrated in *Façade* which takes the player's written input and interprets it into various viable responses [37]. Responses depend on the current story beat being played through, so the same player input can be interpreted in different ways throughout the story (e.g. if the player continually says the word 'no', what they are saying it in response to and how the other characters might respond will vary considerably throughout the narrative). For any given beat, the parser will channel the player's response into one of around eight different interpretations. These can further the current story beat or trigger a new beat. In theory, this approach could negate much of the need for hand-authoring parser interpretations, but in practise the authors ended up creating many ad hoc phrases for specific beats.

5.4 Simulated Agents

While not necessarily a way of reducing labor, creating simulated agents pushes the authorship into a different domain, that of writing patterns of behaviour [3]. With the right tools, this approach can allow the creation of a great deal of novel situations that emerge out of the simulation.

Works like *Prom Week* [40] involve simulating characters, with a narrative coming through the player's interaction with these characters and their simulated behaviour amongst themselves [41]. Martens and Iqbal have made Villanelle, a story-engine for the creation of these kinds of narratives [36]. Among commercial works, agents can be seen in highly procedural games like *Dwarf Fortress*, [1] and more tightly authored experiences like *Elsinore*.

6 Embracing

Rather than seek to design it away, Stern has argued that we ought embrace the combinatorial explosion [61]. The authoring wall is only high in comparison to the capacity of the authors of an interactive narrative. Embracing the burden is the final strategy.

6.1 More Hours

One common method of overcoming the authorial burden is allotting more time to it. This can either mean taking longer to create the work, or more controversially, it can mean crunch: packing more working hours into a short period of time, typically before product launch. While this approach is often baked into the habitus of game creation, extended over-work leads to exhaustion and burnout [14].

6.2 More Writers

One paradigm of interactive narrative is the solo-authored piece, product of a single vision. But the other paradigm is that which is common in videogame companies: to have several writers, often in large teams. While this approach has diminishing marginal returns, if you have more writers, under the right conditions more content can be written.

6.3 Developing Craft Knowledge

Access and education for authors has been highlighted as an important area for developing interactive narrative; [59] being able to think and write interactively is a skill that writers of other media don't automatically possess. The more acquainted authors are with common interactive patterns, and familiar they are with their tools, the more ambitious projects they will be able to complete. It is perhaps no surprise that the most popular narrative text engines, such as Inform 7, Twine and Ink all have extensive documentation.

6.4 Developing Better Tools

It has been argued that improving the 'user experience' of authoring tools can help authors attain greater competency with tools and unlock deeper affordances, [51] as well minimising interruptions to authoring flow [26]. This approach is taken up elsewhere in this volume.

Tool creation itself can be unproductive if the tools are never utilised or re-create the affordances of existing engines. However, for some story-structures, the right kinds of visualisation and testing tools can have a huge effect. Emily Short described such an 'author-friendly toolset for writing Versu stories... that sped up content production by a factor of at least ten and meant that we could produce much bigger, longer stories than previously released' [45].

7 Conclusion

The most basic form of interactive narrative, the pure branching story, creates a large writing burden, sharply curtailing the possible length of such a story. As greater levels of state tracking and content structuring are introduced, the complexity of writing new content increases. Ways of ameliorating this twin burden of the volume of content and difficulty of production can in themselves create new challenges, requiring different skills from authors. Strategies don't necessarily alleviate work, so much as change the form it takes.

When faced with an ambitious idea, the author of an interactive work has many paths to realising it. They may pare back the scope; they may employ structures like merging and loops to keep branching under control; they could make use of more advanced forms, using salience to decide what to display, or procedurally generating a large number of variations. They might take their time, improve their skills or work with others to see the idea through.

The authorial burden then isn't a hard limit on what can be achieved, but a malleable border that shifts. It is the point at which an author is willing to compromise between their vision, and what tools, time and their own powers allow them to achieve.

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